

Course “Theoretical Ecotoxicology: Concepts and Application”

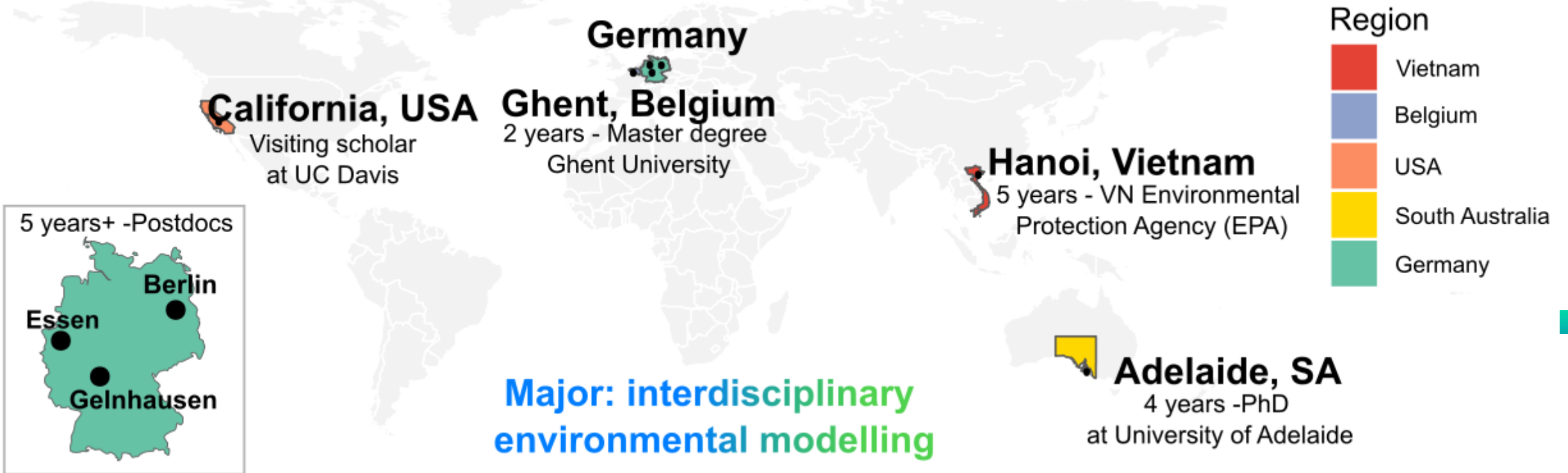
Dealing with complex data and designs (incl. Mixed effect models)

Hanh H. Nguyen & Verena C. Schreiner

Ecotoxicology

Research Center One Health Ruhr
& University of Duisburg-Essen

Research travel journey

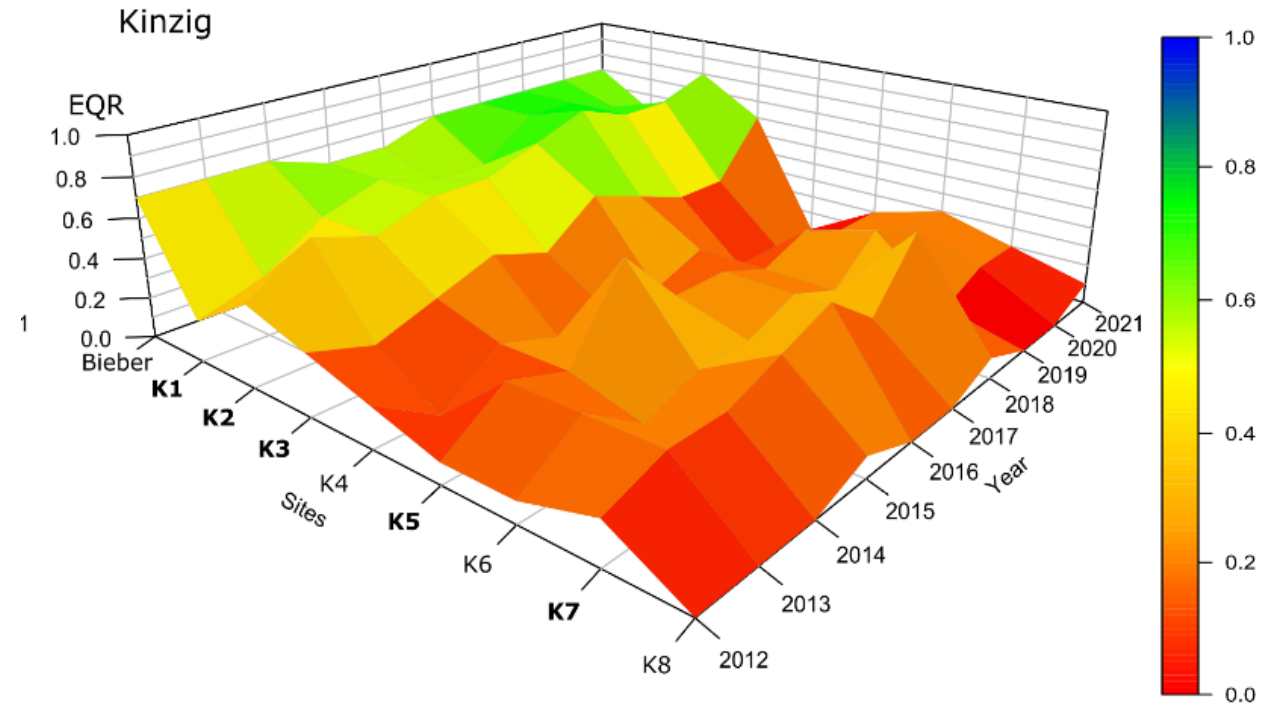


- PhD: Modelling effects of climate, land-use, and nutrients on diatoms and benthic macroinvertebrates in rivers and reservoirs
- Postdocs: Environmental – Eco(toxico)logical modelling

INTRODUCTION – HANH NGUYEN

Current research projects:

- Trend analyses of in-stream & non-target screening substances: Rhine, Elbe, Danube (2005-2021).
- Trend analyses of aquatic communities: LTER sites (Boye & Kinzig) and German-wide scales.
- Trend analyses of cascading multiple stressor interactions on aquatic communities in Europe.



Source: www.umweltprobenbank.de/en

LEARNING OBJECTIVES

- Complex **experimental design** and **data**
- Key experimental design issues
- Data structure, error distribution, and link functions
- Generalized linear models
- Mixed models: fixed and random effects



EXPERIMENTAL DESIGN AND DATA

How are environmental impacts of chemicals assessed?



Which ecotoxicological assessments do you know?



Which data are collected?



EXPERIMENTAL DESIGN AND DATA

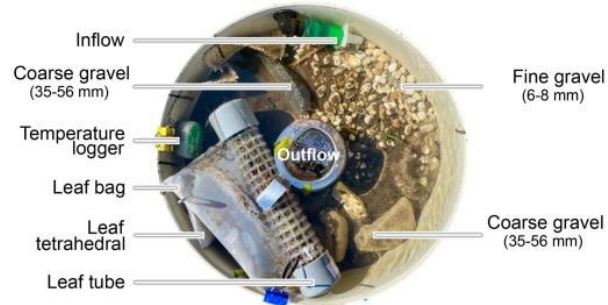
Ecotoxicological assessments

Data

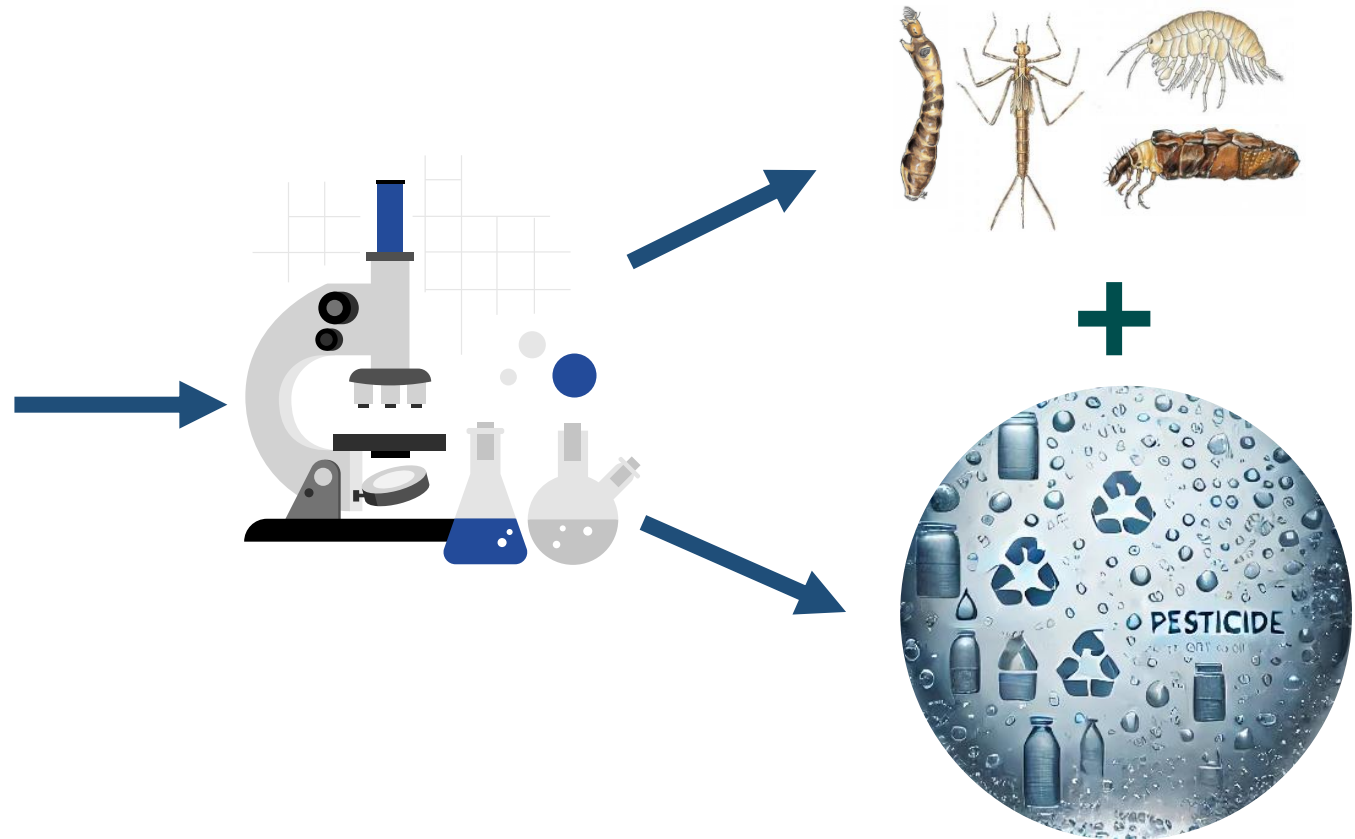
- Ecotoxicity tests



- Mesocosm



- Field sampling



EXPERIMENTAL DESIGN ISSUES

Sample replicates

- Experimental observations with same treatments
- Must be independent ($\neq$ pseudo replication)
- Minimum number of replicates ≥ 3



Sample size (n)

- Number of independent observations to obtain **reliable results**
- Too few samples fail to detect true effects
- Inflated error rates (**Type I and Type II**)



Statistical modeling

- Derive generative conclusions about **significant effects**
- Affected by data structure and experiments (e.g., ANOVA vs linear models vs machine learning)
- Must be included in design phase of study (before the experiments)

SAMPLE SIZE

Data

- Variable X (population mean μ , standard deviation δ)
- Variability of samples: standard error (precision of sample mean estimates of population mean):

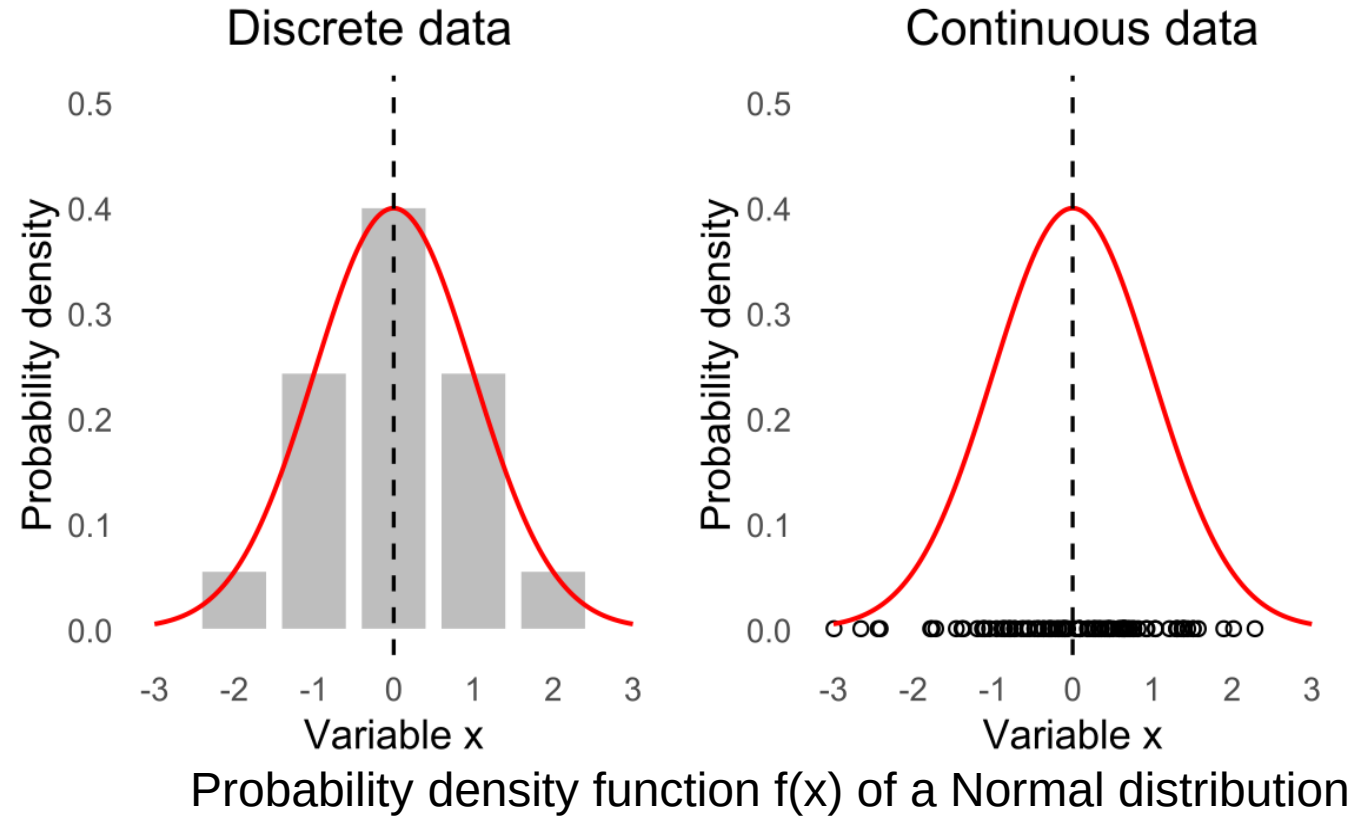
$$\varepsilon = \frac{\text{standard deviation } \delta}{\sqrt{\text{sample size } n}}$$

⇒ Larger n decreases ε , increases statistic power (larger test statistics, lower p -value)

Null hypothesis significant testing (H0)

Error types	Definition
Type I error (false positive)	Rejecting TRUE Null hypothesis
Type II error (false negative)	Failing to reject FALSE Null hypothesis

Small samples n , high chance of failing to detect true effects



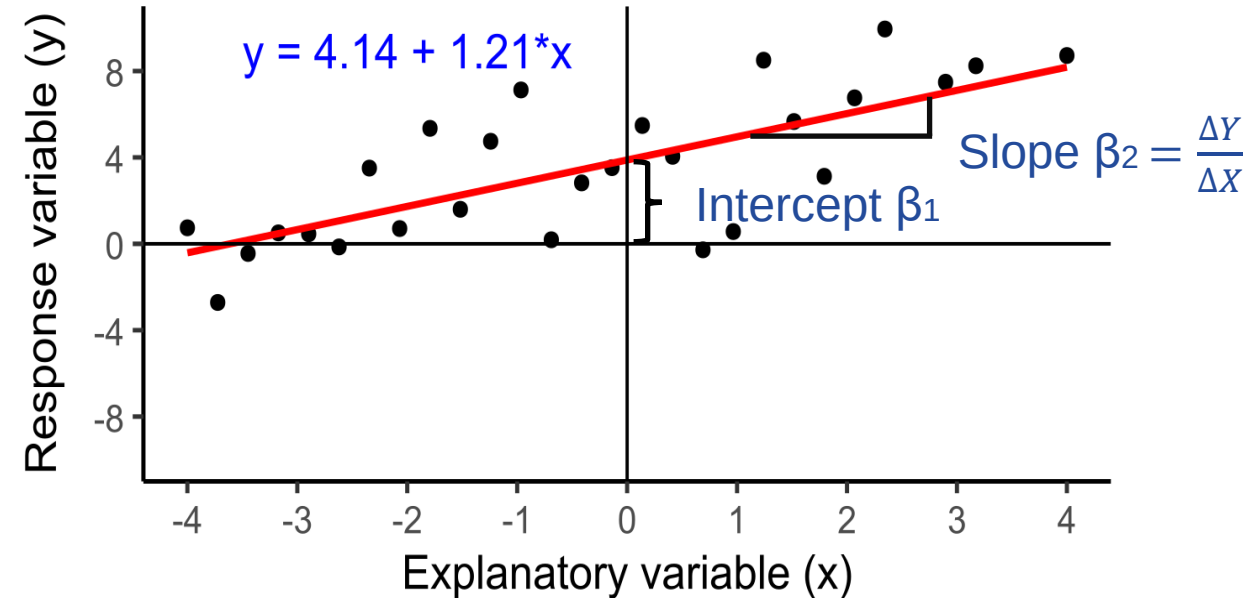
LINEAR MODELS

Simple linear regression model:

$$Y = \beta_1 + \beta_2 X + \varepsilon$$

- Y – response (dependent) variable
- X – explanatory/predictor (independent) variable
- β_1 , β_2 – intercept and slope coefficients
- ε – standard residual error (unexplained variance)

Simple linear regression



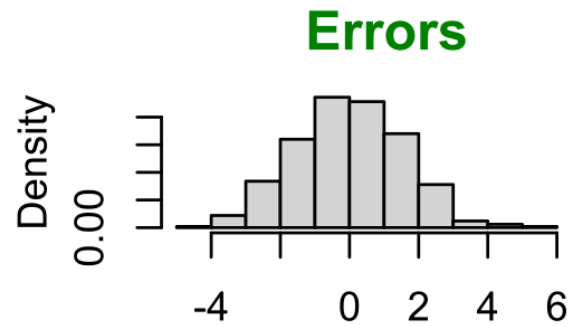
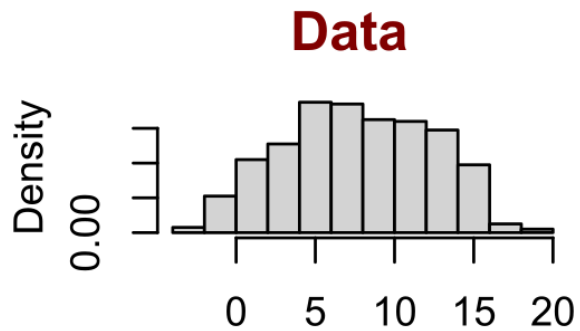
Key model assumptions:

- Linear relationships $y \sim x$
- Normality (of error terms)
- Homoscedasticity (constant variance of residuals)
- No outliers/ Independence of observations (residuals are not correlated)

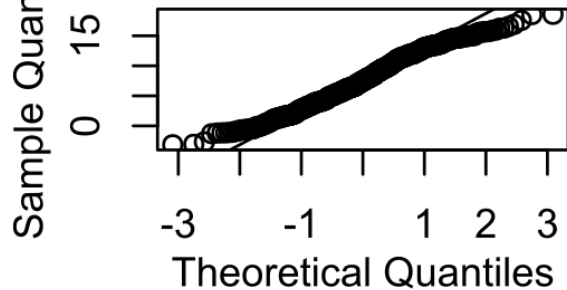
LINEAR MODELS

Key model assumptions:

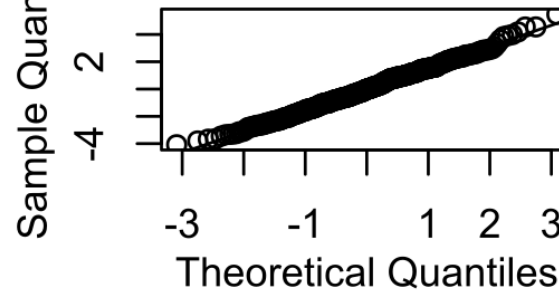
- Normality (of error terms): $\epsilon \sim N(0, \sigma^2)$



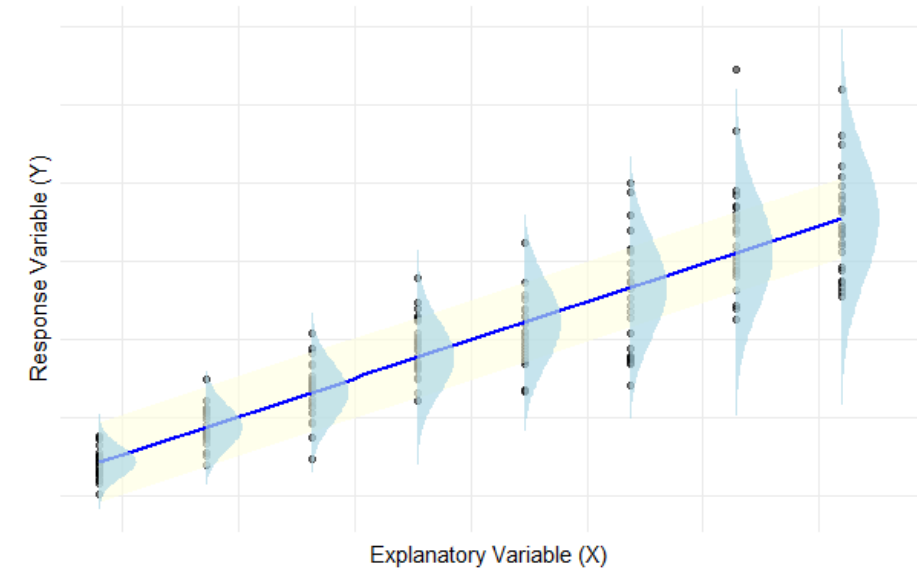
Q-Q plot: Normality of data (wrong!)



Q-Q plot: Normality of errors (right!)



Normal distribution of data



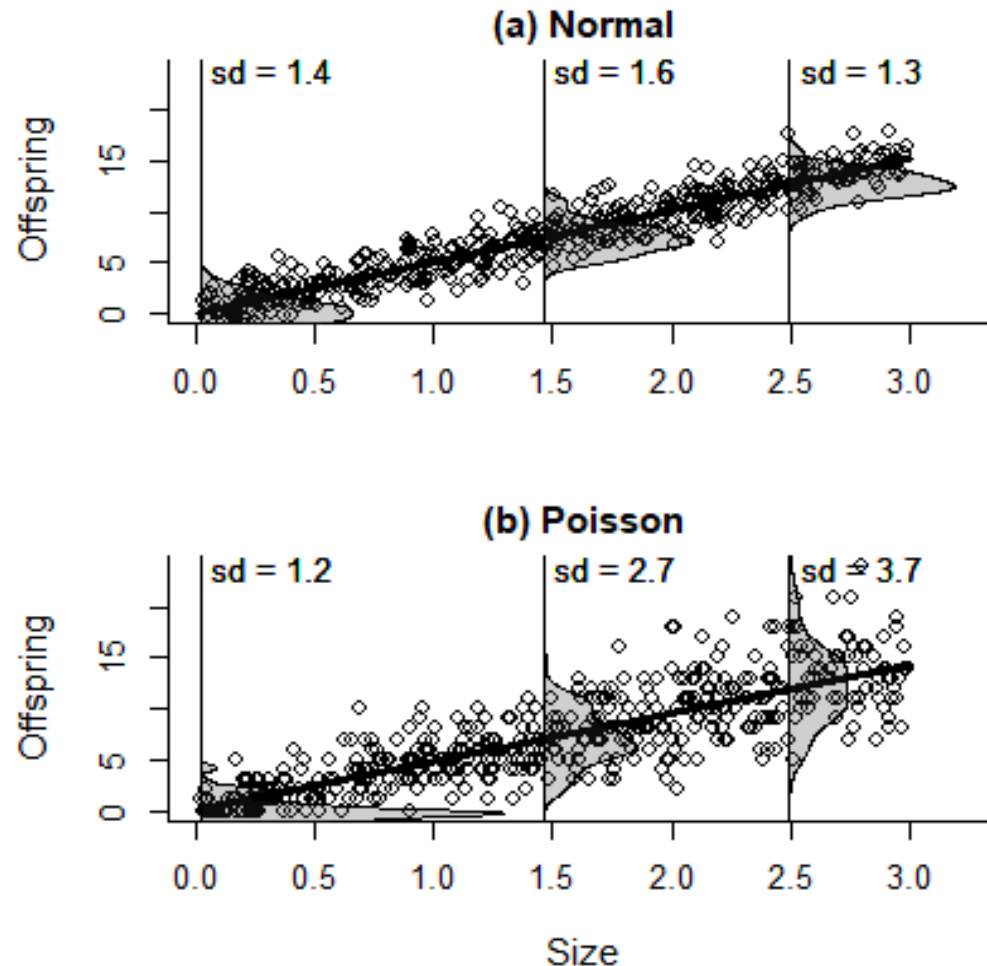
At each value of explanatory variable X , observations of response variable Y are normally distributed

$$Y_i \sim N(\bar{Y}, \sigma_Y^2)$$

LINEAR MODELS

Key model assumptions:

- Homoscedasticity (constant variance of residuals)

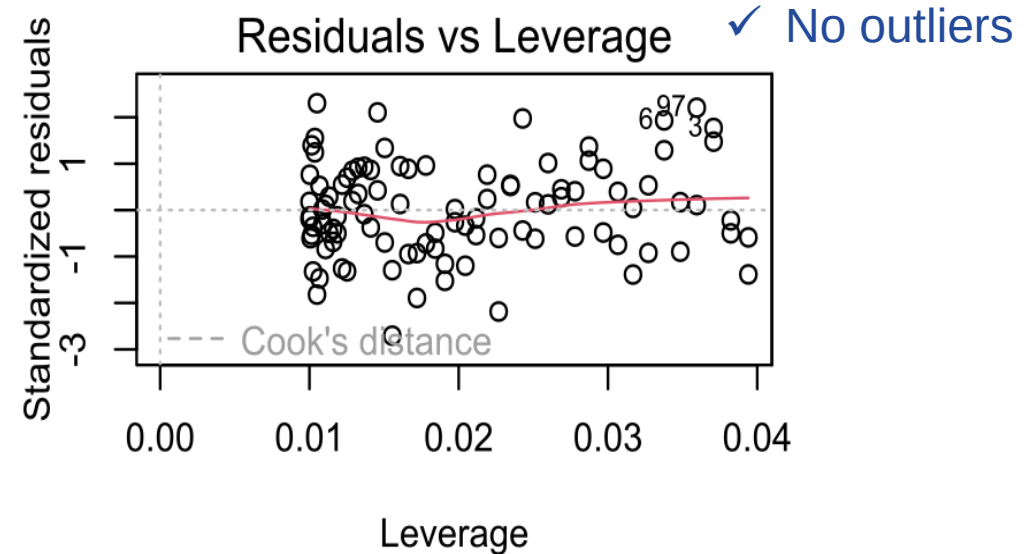
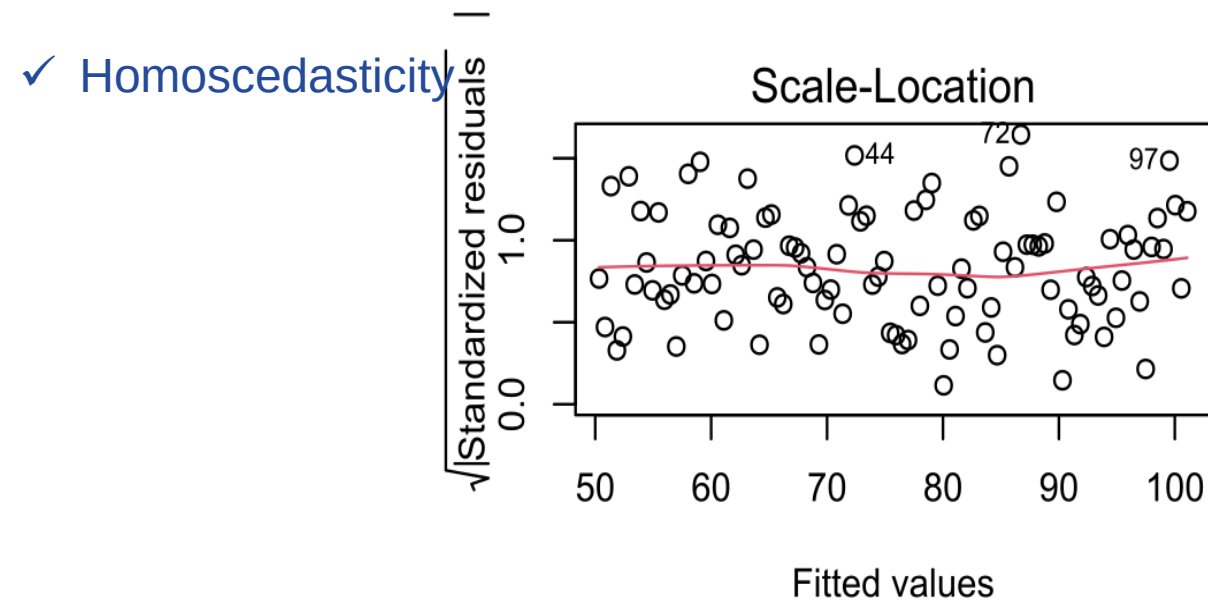
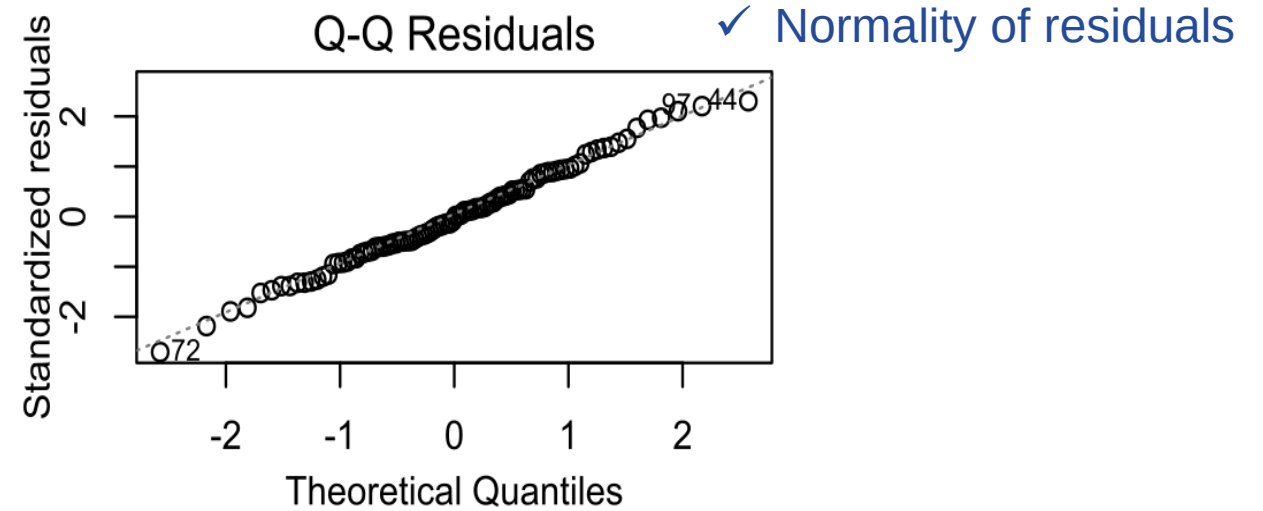
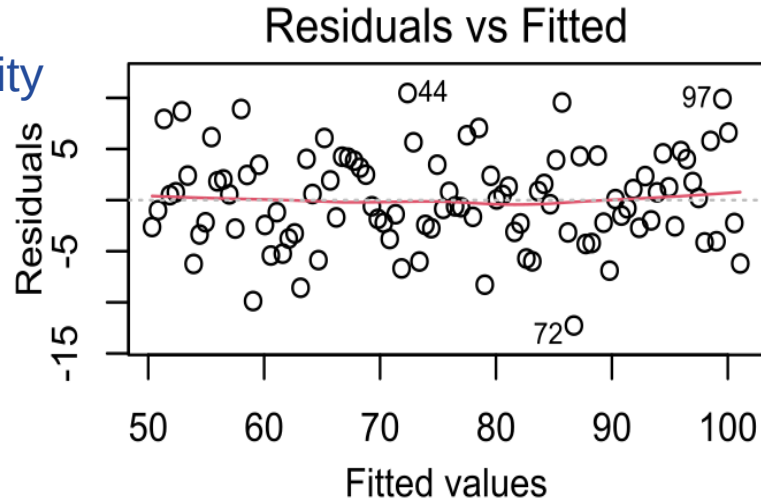


Variance of simulated offspring production (Y) increased with increased mean value of individual body size (X)

~ distribution of $\hat{Y}$ is influenced by both X and ϵ
~ error term ϵ is independent and follows normal distribution (i.e., constant variance)

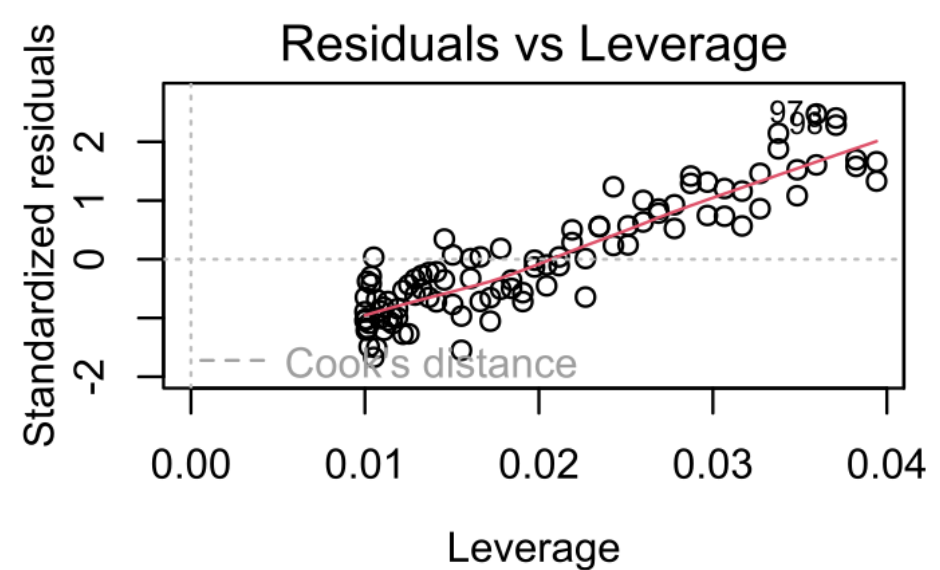
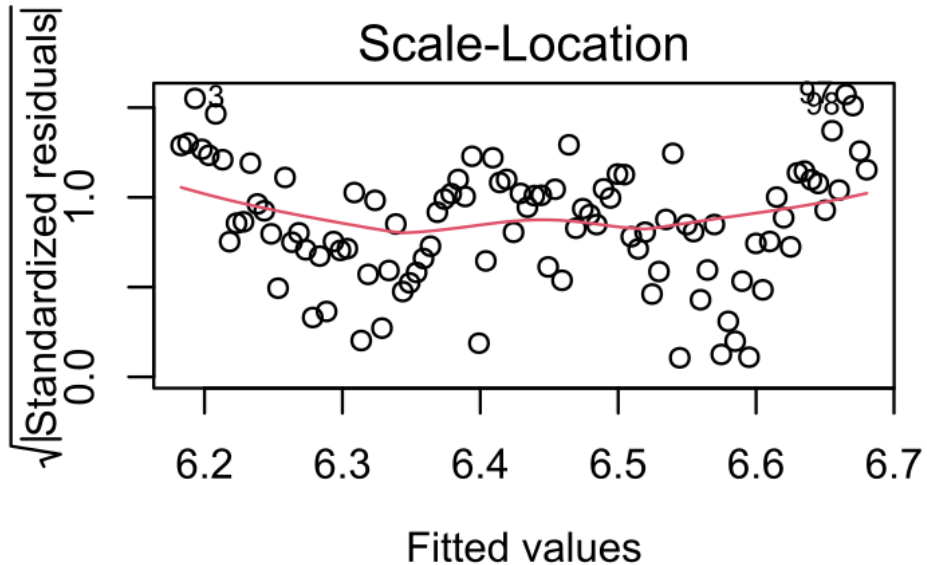
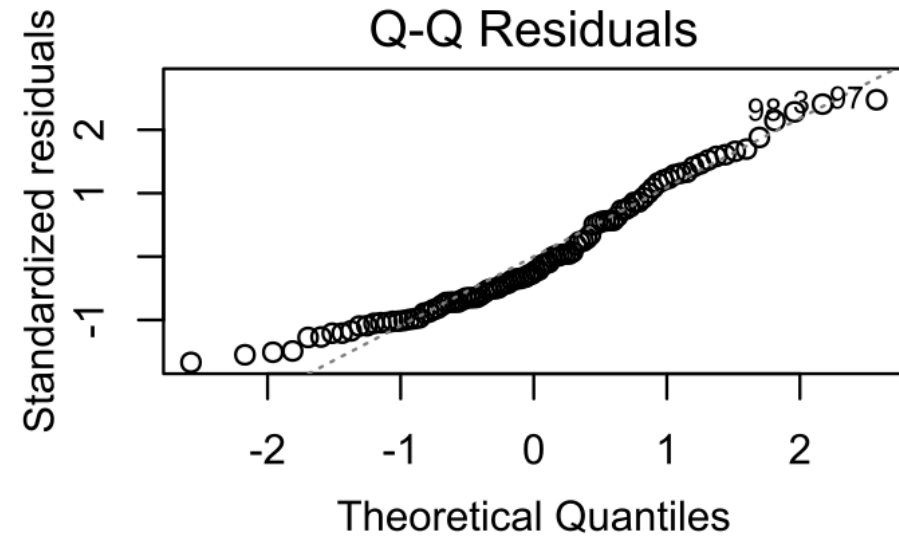
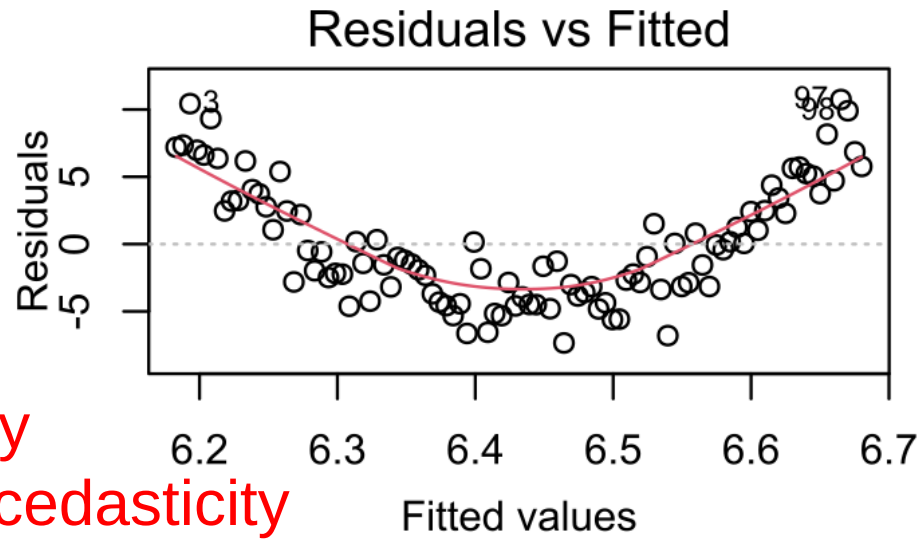
LINEAR MODELS – MODEL VALIDATION – EXAMPLE (1)

- ✓ Linearity
- ✓ Homoscedasticity

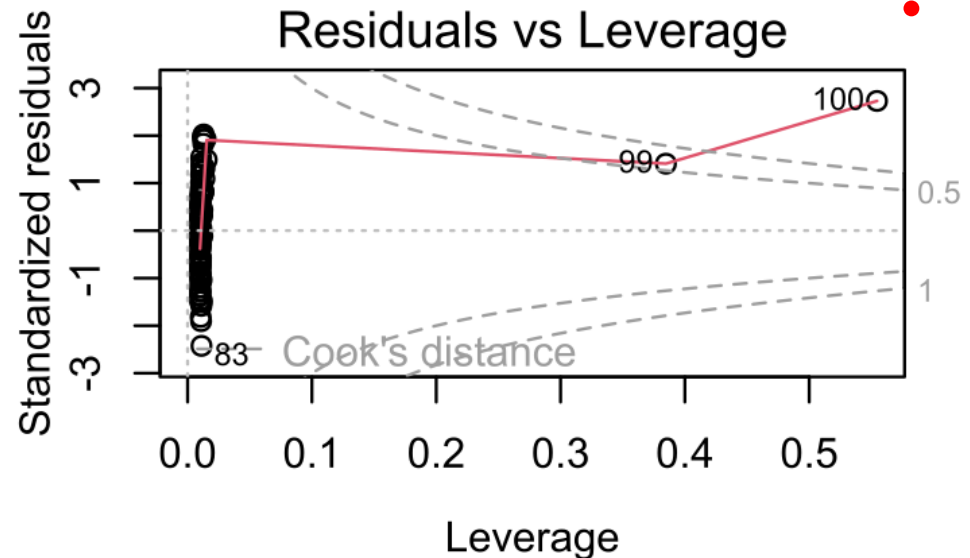
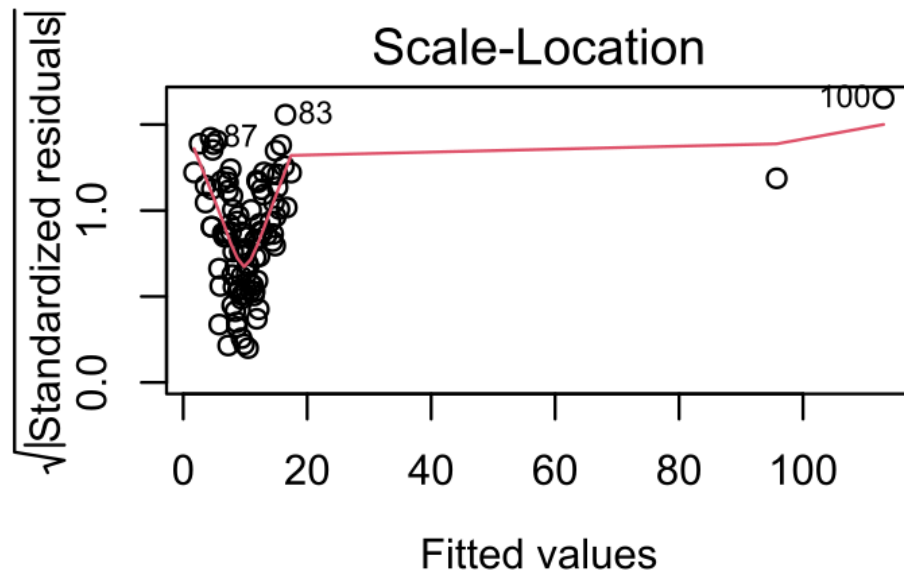
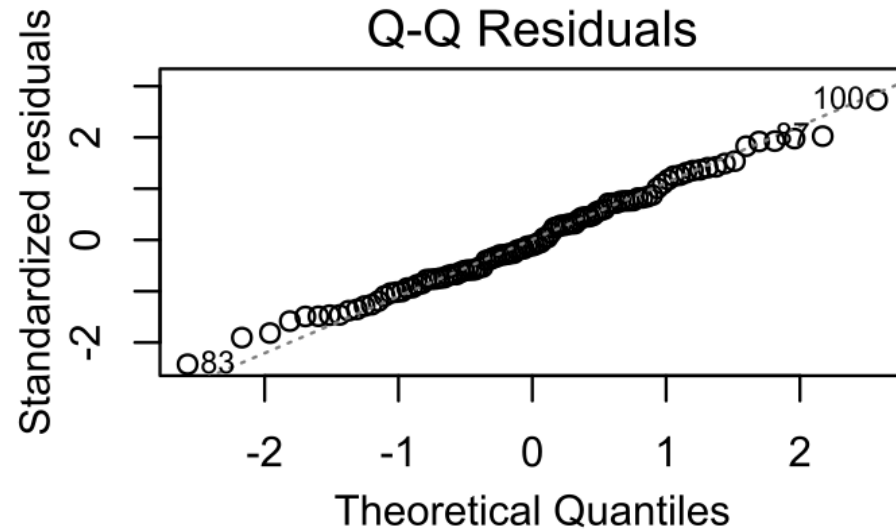
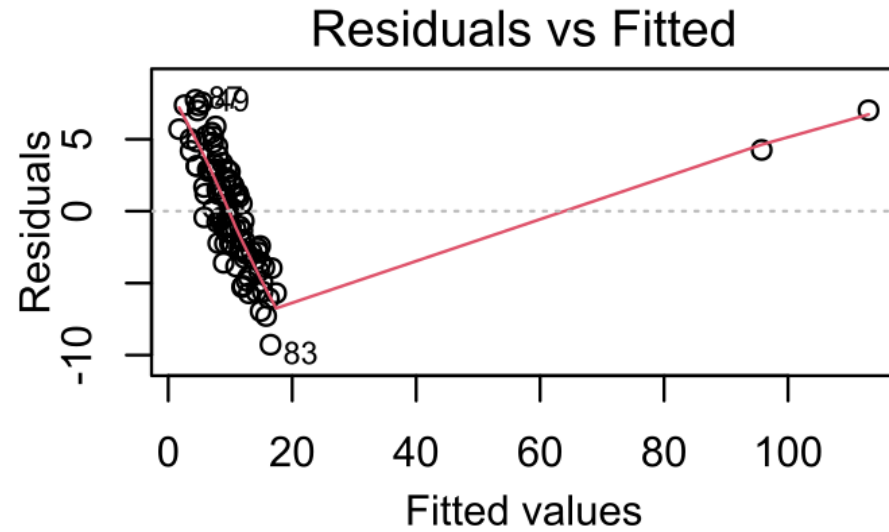


LINEAR MODELS – MODEL VALIDATION – EXAMPLE (2)

- Linearity
- Homoscedasticity

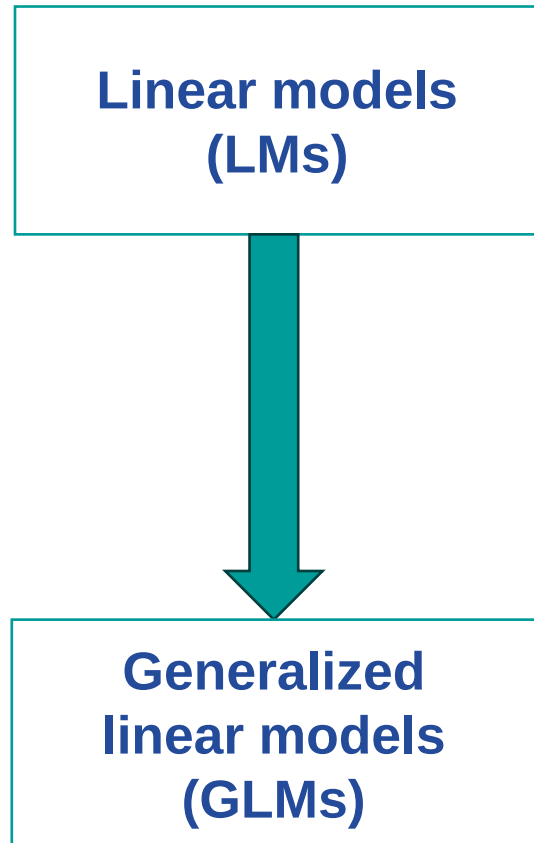


LINEAR MODELS – MODEL VALIDATION – EXAMPLE (3)



• Outliers

MODEL CHOICE



LM key assumptions:

- Linear relationships $y \sim x$
- Normality of error terms
- Homoscedasticity (constant variance)
- No outliers/ Independence of observations

GLMs not suitable if:

- Normal error distribution & constant variance
- Complex non-linear relationships
- Dependent observations

GENERALIZED LINEAR MODELS – KEY MODEL COMPONENTS

$$\boxed{Y} = \boxed{\beta_1 + \beta_2 X} + \boxed{\varepsilon} = \beta X + \varepsilon$$

Linear predictor $\eta = \beta X$

- Equivalent to simple linear regression model
- Linear predictor η is predicted by explanatory variable(s)

Error distribution ε

- Represents part of Y variation not explained by linear predictor η
- Variance $\text{var}(Y)$ depends on the mean $\mu \Rightarrow$ Selection based on response variable type Y

Link function $g(\mu)$

- Connect expected mean of the response variable $E(Y|X) = \mu$ and linear predictor η
- For LM with normal error distribution $\varepsilon \sim N(\mu, \delta^2)$, the link function is the identity link

GENERALIZED LINEAR MODELS – EXPONENTIAL FAMILY OF DISTRIBUTIONS

Error distributions, link functions for commonly modelled response variables

(Source: Adapted from Buckley, 2015; Crawley, 2012)

Response Variable	Error Distribution	Default Link	Alternative Links	Link Equation	Variance Function	Example
Continuous (positive and negative values)	Normal	Identity	Log, Inverse	$\eta = \mu$	Constant	Weight gain Abundance (log link)
Positive continuous	Gamma	Inverse	Log, Identity	$\eta = 1/\mu$	μ^2	Time to germination / death
Continuous, proportion 0-1 (not including 0 and 1)	Beta	Logit	Probit, Log, Identity	$\eta = \log(\mu / (1 - \mu))$	$\mu(1 - \mu)$	Fraction of dead organisms Ratio of hatching success
Count (non-negative integers)	Poisson	Log	Identity, Sqrt	$\eta = \log(\mu)$	μ	Taxa richness Number of dead organisms
Count (with overdispersion)	Negative Binomial	Log	Identity, Sqrt	$\eta = \log(\mu)$	$\mu + \kappa\mu^2$ (κ = shape)	Parasites per organism
Proportion or binary (0/1)	Binomial (binary, proportion)	Logit	Probit, Log, Identity	$\eta = \log(\mu / (1 - \mu))$	$\mu(n - \mu) / n$	dead / alive male / female

Interested in other error distributions? Type “?family” in R

GENERALIZED LINEAR MODELS – MAXIMUM LIKELIHOOD ESTIMATION (MLE)

Likelihood (L):

- A function that is proportional to the probability of generating the data (given the model):

$$L(\phi | \mathbf{Y}) = f(\mathbf{Y} | \phi)$$

$f(\mathbf{Y} | \phi)$ is the probability density function of continuous data
(or probability mass function of discrete data) given the parameters ϕ

- Likelihood is Gaussian in LMs, and likelihood function adapts to error distributions in GLMs.



Maximum likelihood estimation (MLE)

- Method of maximizing likelihood function (L) by estimating model parameters that most likely generated the data.
- In GLMs to find the MLE requires iterative numerical methods to find the MLE and associates with specific error distributions.

GENERALIZED LINEAR MODELS – MEASURE OF MODEL FIT

Linear model

- **Sum square of errors (SSE):** measures total squared difference between observed response variable (Y_i) and model predictions ($\hat{\mu}_i$): $SSE = \sum_{i=1}^n \epsilon^2 = \sum_{i=1}^n (Y_i - \hat{\mu}_i)^2$
- Goodness-of-fit: $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{\mu}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$
- Minimizing SSE using ordinary least square is equivalent to obtaining maximum likelihood for model parameters under the assumption of normal distribution.



Generalized linear model:

- **Deviance (D):** measures deviance of fitted GLM with a perfect/saturated model (fitted responses $\hat{\mu}_i$ equal observed responses Y_i): $D = 2\Phi(l_{sat} - l_{mod})$
where: Φ is dispersion parameter, l_{mod} is the log-likelihood of the fitted model,
 l_{sat} is the log-likelihood of the saturated model
- Goodness-of-fit: $PseudoR^2 = 1 - \frac{Residual\ deviance}{Null\ deviance}$
- Minimizing Deviance using iterative algorithms (e.g., reweighted least squares IWLS, Newton-Raphson, or Fisher Scoring) is equivalent to MLE under the assumption of exponential family distribution.

GENERALIZED LINEAR MODELS - EXAMPLE

Binomial distribution: Study the probability of wild boar having **Tuberculosis (Tb)** as a function of the animal **Length** (i.e., length from nose to the tail joint along the back of the animal, in cm)

```
# Reference: Zuur et al. (2009). Mixed Effects Models and Extensions in Ecology with R
# Data source: Vicente et al. (2006)
# www.highstat.com
```

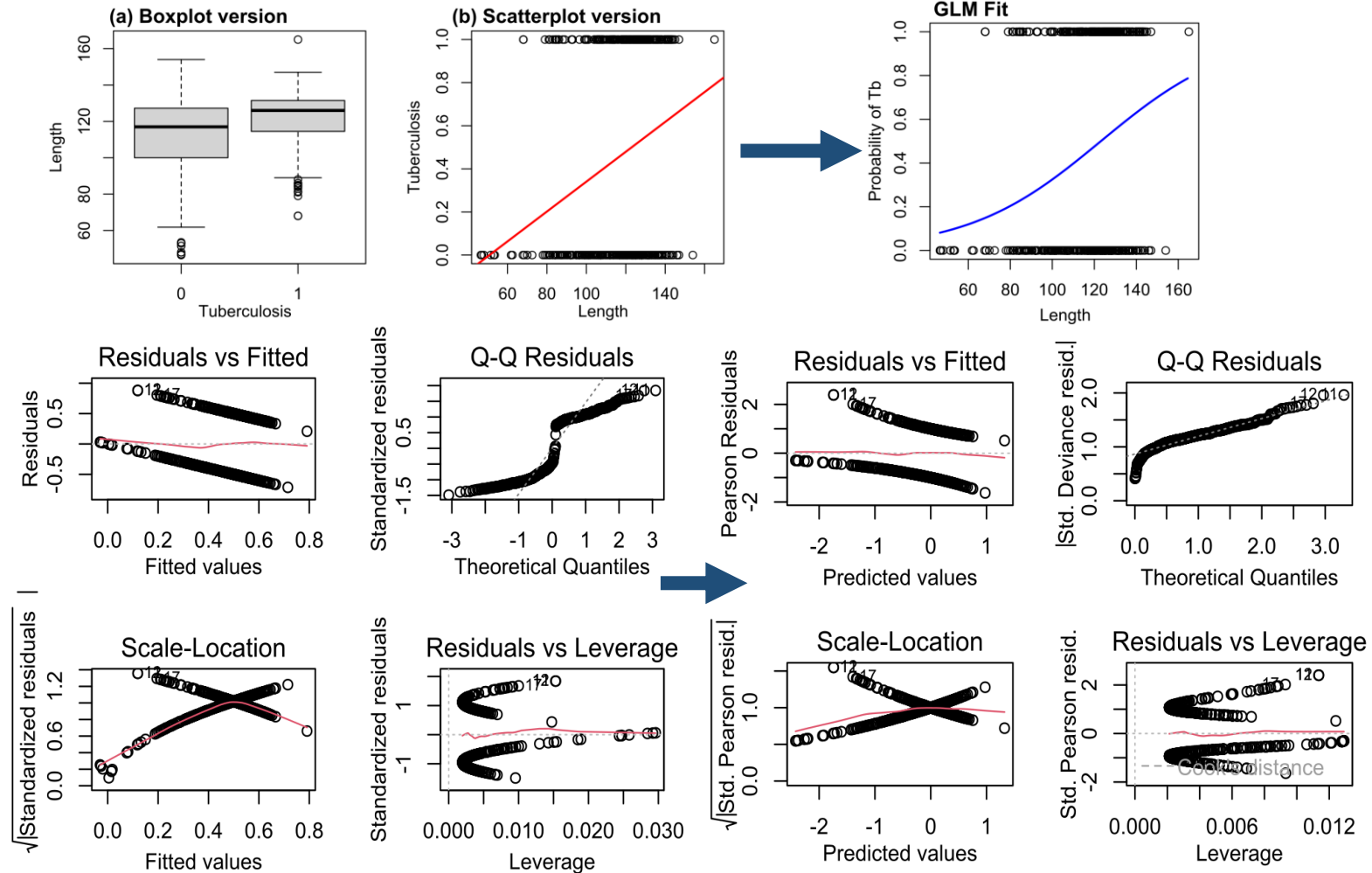
```
#####
## Load data
Tb_data <- read.table("Boar.txt", header = TRUE, sep = "\t")
```

```
#####
## Linear regression:
lm_model <- lm(Tb ~ LengthCT, data = Tb_data_clean)
```

```
summary(lm_model)
```

```
par(mfrow = c(2, 2))
plot(lm_model)
par(mfrow = c(1, 1))
```

```
#####
## Generalized linear regression:
# Model
glm_model <- glm(Tb ~ LengthCT, family = binomial, data = Tb_data_clean)
# Model summary
summary(glm_model)
# Model validation plots
par(mfrow = c(2, 2))
plot(glm_model)
par(mfrow = c(1, 1))
```



GENERALIZED LINEAR MODELS - EXAMPLE

```
# Reference: Zuur et al. (2009). Mixed Effects Models and Extensions in Ecology with R
# Data source: Vicente et al. (2006)
# www.highstat.com
```

```
#####
## Load data
Tb_data <- read.table("Boar.txt", header = TRUE, sep = "\t")
```

```
#####
## Linear regression:
lm_model <- lm(Tb ~ LengthCT, data = Tb_data_clean)
```

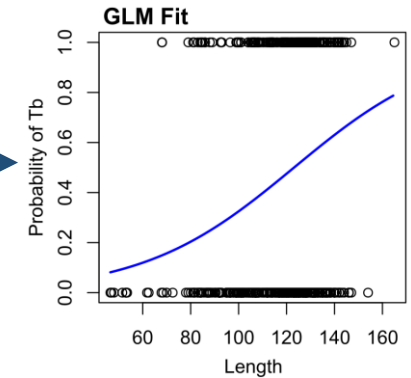
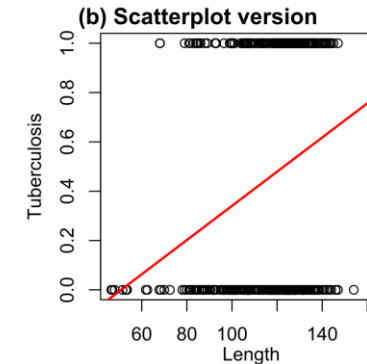
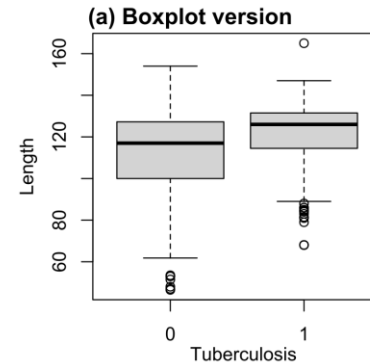
```
summary(lm_model)
```

```
par(mfrow = c(2, 2))
plot(lm_model)
par(mfrow = c(1, 1))
```

```
#####
## Generalized linear regression:
```

```
# Model
glm_model <- glm(Tb ~ LengthCT, family = binomial, data = Tb_data_clean)
```

```
# Model summary
summary(glm_model)
# Model validation plots
par(mfrow = c(2, 2))
plot(glm_model)
par(mfrow = c(1, 1))
```



```
lm(formula = Tb ~ LengthCT, data = Tb_data_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-0.7148 -0.4710 -0.2262  0.4723  0.8812

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.352524   0.134628  -2.619   0.0091 **
LengthCT     0.006931   0.001136   6.103 2.07e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4819 on 506 degrees of freedom
Multiple R-squared:  0.06856, Adjusted R-squared:  0.06672
F-statistic: 37.24 on 1 and 506 DF, p-value: 2.075e-09
```

```
glm(formula = Tb ~ LengthCT, family = binomial, data = Tb_data_clean)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.892109   0.671152  -5.799 6.67e-09 ***
LengthCT     0.031606   0.005588   5.656 1.55e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 700.76  on 507  degrees of freedom
Residual deviance: 663.56  on 506  degrees of freedom
AIC: 667.56

Number of Fisher Scoring iterations: 4
```

$$\text{logit}(\pi) = -3.89 + 0.03 * \text{LengthCT}$$
$$\pi = 1 / (1 + \exp(-\text{logit}(\pi)))$$

Examples: LengthCT = 60 => p = 0.119
LengthCT = 140 => p = 0.630

GENERALIZED LINEAR MODELS - EXERCISES

- Exercise 1 - Binomial GLM
- Exercise 2 - Poisson GLM
- Exercise 3 - Negative Binomial GLM

```
Call:
glm(formula = TOT.N ~ D.PARK, family = poisson, data = RK_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	4.316e+00	4.322e-02	99.87	<2e-16	***
D.PARK	-1.059e-04	4.387e-06	-24.13	<2e-16	***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 1071.4 on 51 degrees of freedom
Residual deviance: 390.9 on 50 degrees of freedom
AIC: 634.29

Number of Fisher Scoring iterations: 4

$$\begin{aligned} PseudoR^2 &= 1 - \frac{Residual\ deviance}{Null\ deviance} \\ &= 1 - (390.9/1071.4) = 0.635 \end{aligned}$$

Dispersion refers to variability/spread of data points around the expected value predicted by a model.

Overdispersion occurs when random variables have greater variance than expected by assumed distribution.

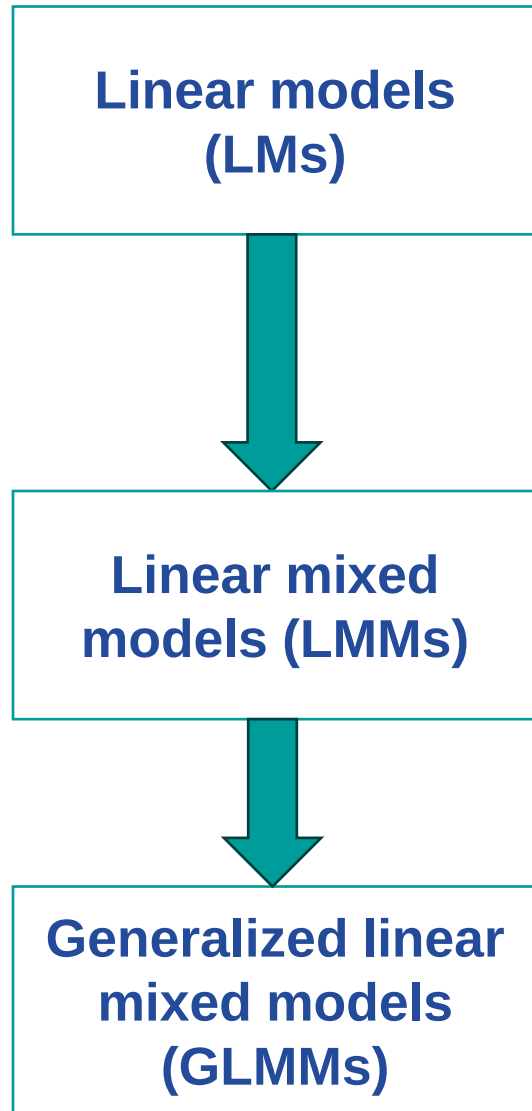
Null deviance: refers to the null model, in which all of the terms are excluded, except the intercept. The degrees of freedom are the sample number $n - 1$.

Residual deviance: refers to the fitted model, which has $n - p$ degrees of freedom, where p is the number of parameters, including the intercept.

AIC: measure of fit that penalizes for the number of parameters. Smaller AICs indicate better model fit.

Fisher scoring iterations: number of iterative steps to find parameter estimates that maximize likelihood function. Smaller values indicate fast convergence.

MODEL CHOICE



LM key assumptions:

- Linear relationships $y \sim x$
- Normality of error terms
- Homoscedasticity (constant variance)
- Independence of observations

LMM is a specific case of GLMM:

- Normality of error terms
- Non-independent or hierarchical data
(repeated measurements/replicates, observations clustered within groups)
~ **Pseudoreplication**

LINEAR MIXED MODELS – KEY MODEL COMPONENTS

Mixed models

- **Fixed effects:** effects of main predictor variables (population-level relationships)
- **Random effects:** effects for different levels of categorical grouping variables (group-level variability, including within σ^2_{within} and among σ^2_{among} groups)

Mixed model form (in matrix notation):

$$Y = X\beta + \epsilon + Zb$$

where: Y – vector ($n \times 1$) of a response variable

X – matrix ($n \times p$) of p predictor/explanatory variables

β – vector ($p \times 1$) of fixed-effects regression coefficients

$\epsilon \sim N(0, R)$ – vector ($n \times 1$) of residuals (i.e., part of y not explained by the model)

Z – matrix ($n \times q$) of q random effects

$b \sim N(0, G)$ – vector ($q \times 1$) of random effects (i.e., deviation around the fixed β)

When to treat a predictor variable as a random or a fixed effect?

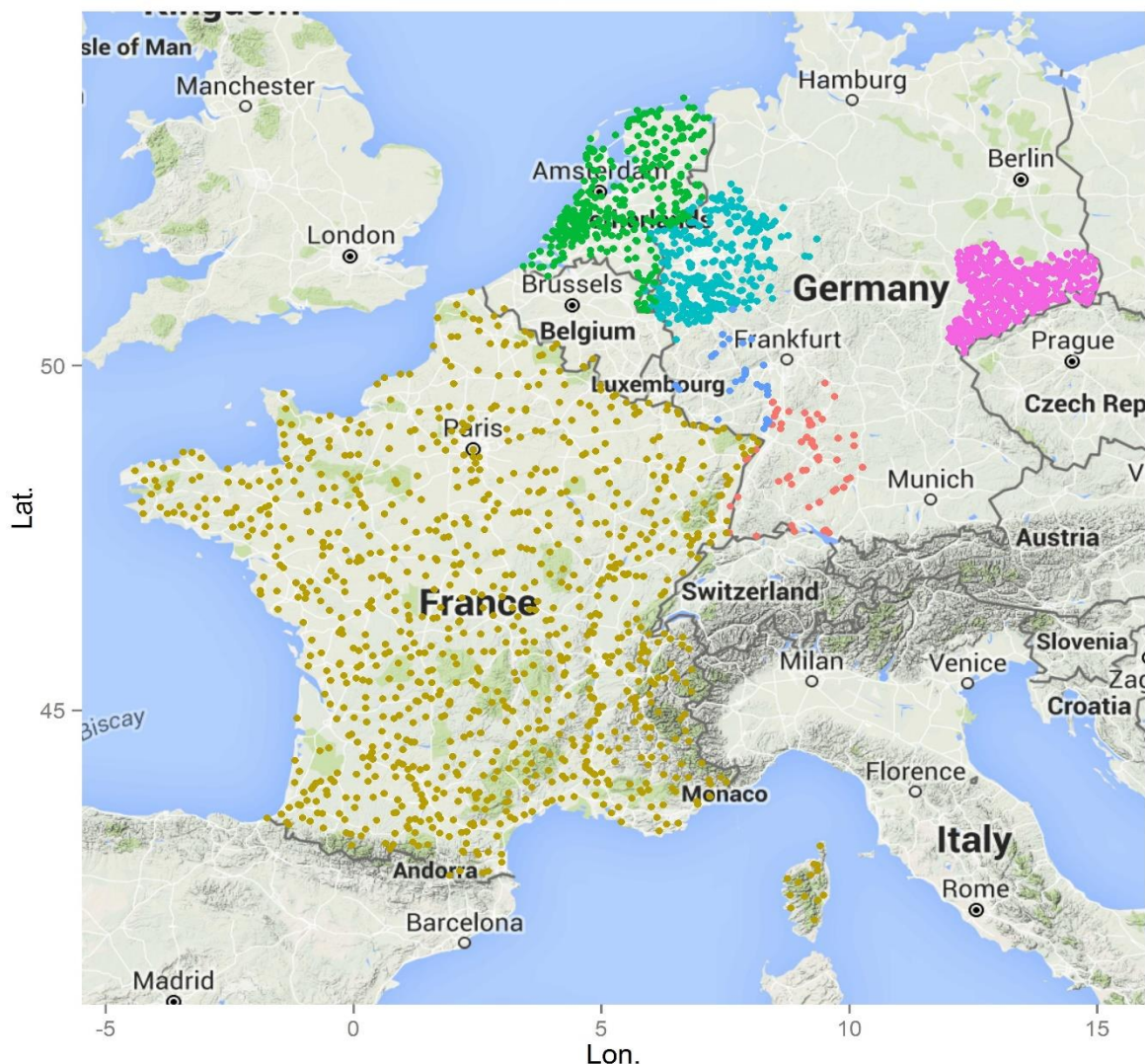
LINEAR MIXED MODELS – EXAMPLE (1)



- Response is recorded per organism
→ Organisms are not independent, are part of a group

Model: $\text{response} \sim \text{explanatory} + (1|\text{beaker})$

LINEAR MIXED MODELS – EXAMPLE (2)



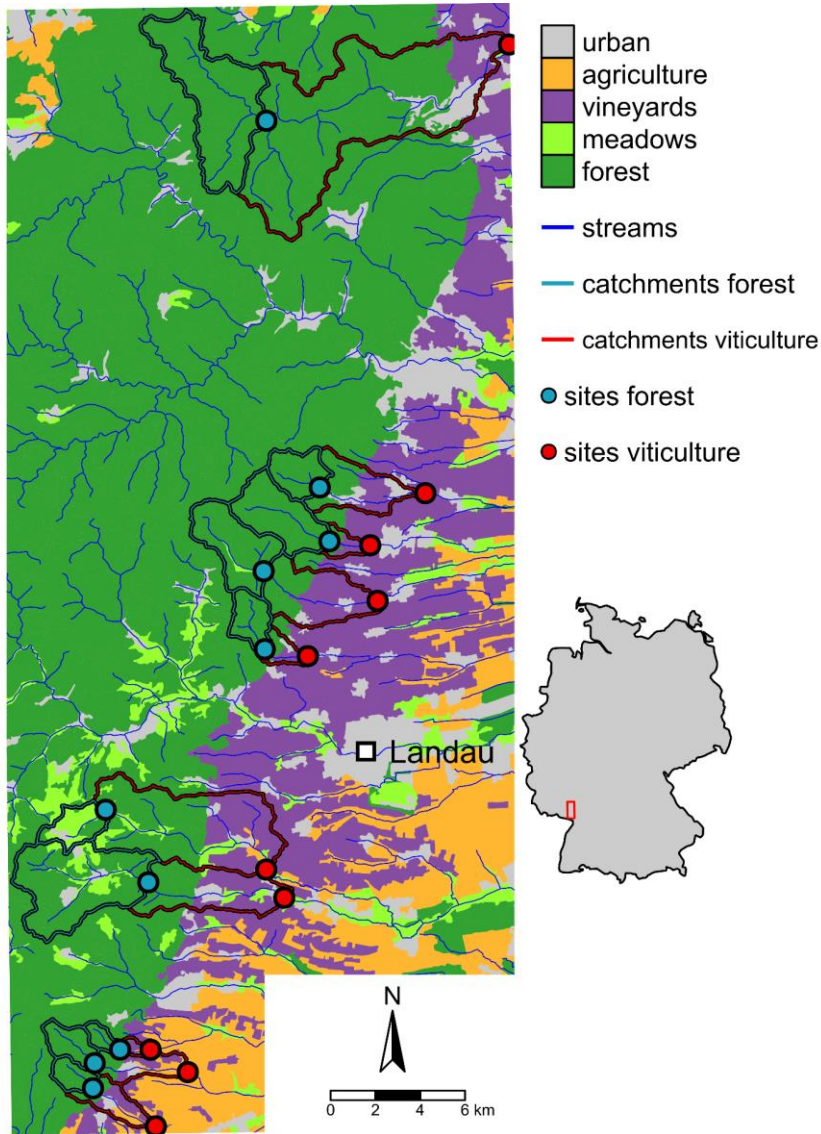
- Pesticide presence/absence data
- Data from 3 countries / 6 regions

→ Monitoring programs might differ by county / region

Model:
 $\text{response} \sim \text{explanatory} + (1|\text{region})$

Source: Schreiner, V.C., et al., 2016. Science of The Total Environment

LINEAR MIXED MODELS – EXAMPLE (2)



- 10 streams, each with two sites
- 4 time points
- 7 replicates per site

→ Repeated measurements

→ Streams are replicates, in-site only pseudo-replicates

Model:

response $\sim$ time_point + site + (1|stream) + (1|site_time_point)

Source: Schreiner, V.C., et al., 2025. Global Change Biology

LINEAR MIXED MODELS – RANDOM EFFECTS

Use of random effects if (Source: Crawley 2002, Gelman 2005):

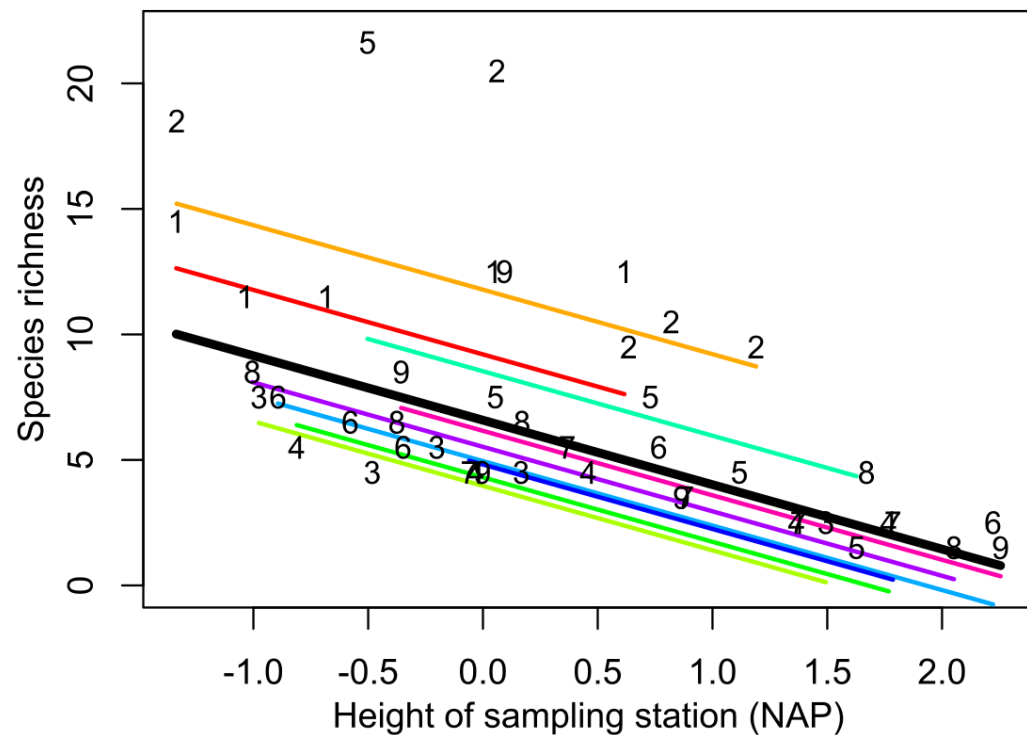
- Have a categorical predictor that is not of direct interest (but its effect should be controlled)
- Interested in quantifying variability among levels of grouping variable
- Interested in combining information across levels of grouping variable
- Not interested in testing hypotheses about differences between responses at specific levels of grouping variable

If grouping variable has fewer than five levels, variable should be treated as a fixed predictor (i.e., variance from a very small sample is unstable)

LINEAR MIXED MODELS – SIMPLE VS. COMPLEX RANDOM EFFECTS

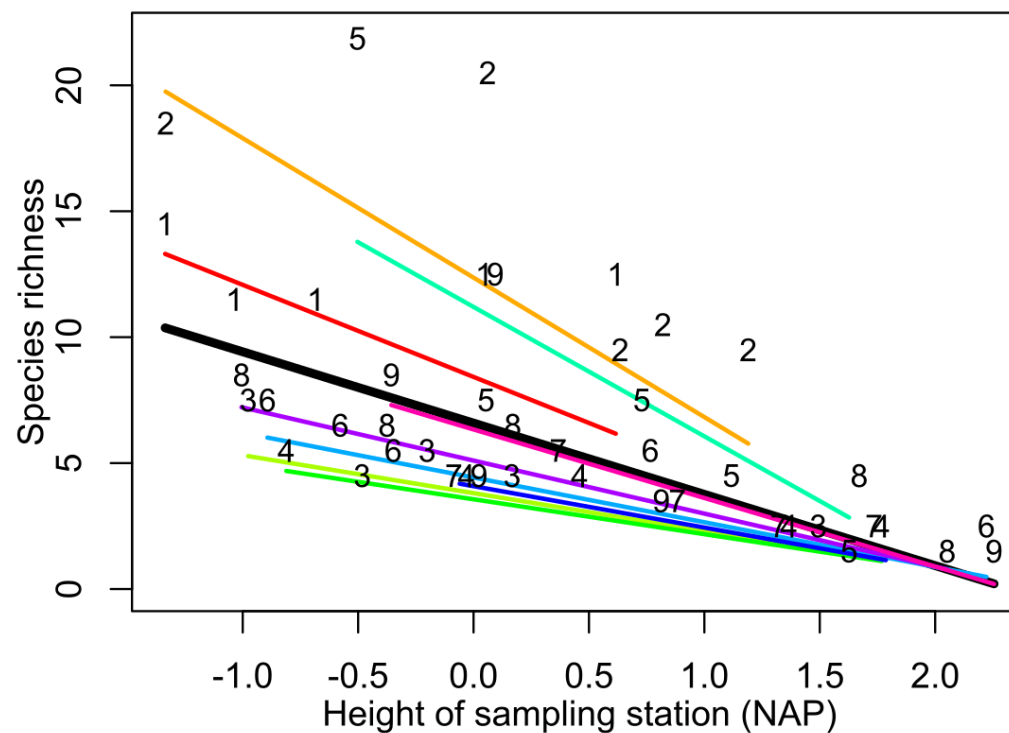
Random intercept only:

- Intercept changes per grouping variable
- Species richness $\sim$ NAP + (1 | fBeach)



Random intercept and slope:

- Includes fixed x random effect interactions
- Species richness $\sim$ NAP + (1 + NAP | fBeach)



Linear mixed model of marine benthic species richness at nine inter-tidal area (fBeach). Thick black line represents the fitted values for the population. Numbers represent the beach grouping variable.

LINEAR MIXED MODELS – CROSSED VS NESTED RANDOM EFFECTS

Interactions between random effects?

Crossed effects

- Levels of one effect group appear in multiple levels of another effect group
- Example: Species are measured as multiple sites and multiple years
 - Sites and years are crossed: variation among sites applies across years, variations among years applies across sites, and variation among site-by-year combinations.
 - Model expression: $(1|\text{site}) + (1|\text{year})$

Nested effects

- Each member of lower-level group belongs to only one higher-level group
- Example: Chick nested within Brood nested within Site (Source: Bolker, 2015)
 - Effects are nested: Chick occurs in only one brood, and each brood occurs in only one site.
 - Model expression: $(1|\text{Chick}/\text{Brood}/\text{Site})$ or equivalent $(1|\text{Site}) + (1|\text{Site}:\text{Brood}) + (1|\text{Site}:\text{Brood}:\text{Chick})$

LINEAR MIXED MODELS – RESTRICTED MAXIMUM LIKELIHOOD (REML)

Marginal Likelihood:

- Marginal likelihood is a likelihood used in MLE for mixed models.
- Marginal likelihood is a modified form of likelihood that integrates random effects b :

$$L_{\text{marginal}}(\beta, \phi | Y) = \int L(\beta, b | Y) * L(b | \phi) db$$

ϕ is the among-group variance components (random effects covariance)

Marginal likelihood of observed data Y is the integral of the conditional likelihood (i.e., $L(\beta, b | Y)$) weighted by the likelihood of random effect b (which is $L(b | \phi)$)

Restricted (REML) likelihood

- REML is a statistical method enabling more accurate estimates of variance components (e.g., random effect variance, error variance) in mixed models.
- REML can be seen as a marginal likelihood, with fixed effects integrated out (e.g., group means) and then variance is estimated from the residuals (part of data not explained by fixed effects).

LINEAR MIXED MODELS - EXAMPLE

Study of factors affecting growth of White Stock
(*Ciconia ciconia*) nestlings in eastern Algeria
(Data source: Bouriach et al., 2015)

```
## Load packages
library(lme4)      # lmer()
library(nlme)      # lme()

## Load data
WS <- read.csv(file = "White_stork_Growth_20112012_V2.csv", header = TRUE)
WS1 <- subset(WS, Year == 2012)
WS2 <- na.exclude(WS1[, c("Beak", "Age", "Nest", "Chick")])
WS2$fNest <- factor(WS2$Nest)
WS2$fChick <- factor(WS2$Chick)

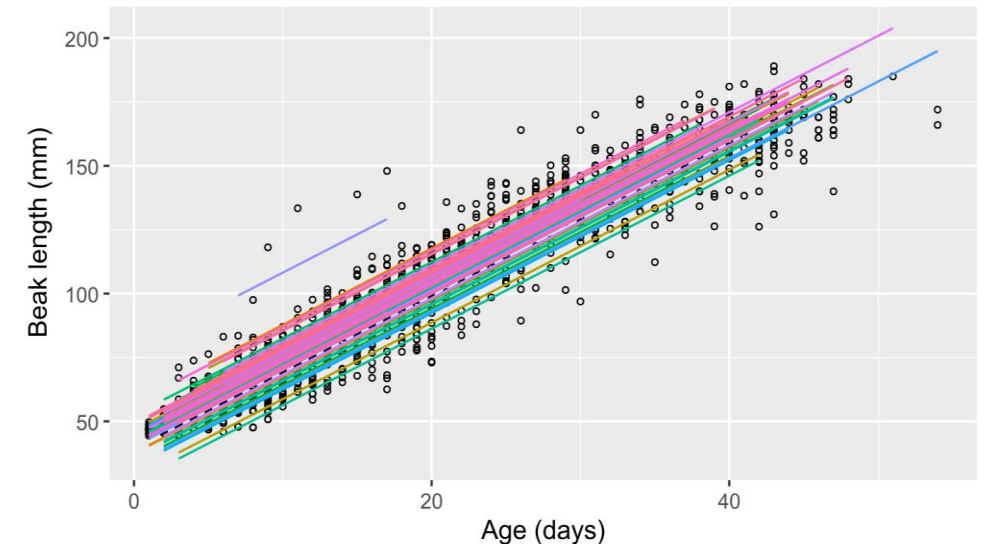
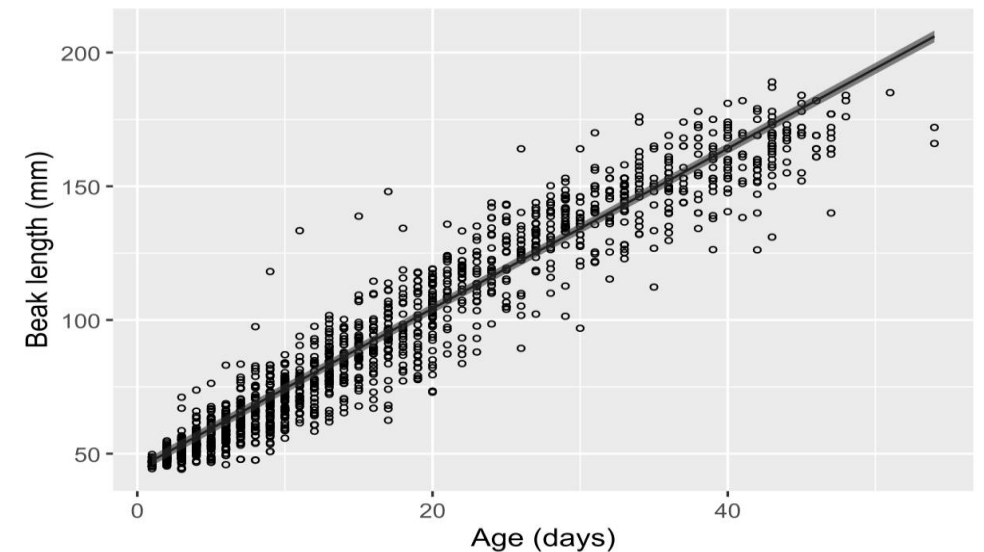
## Models
# Model 1 - Wrong model: fNest as a random effect (ignoring random effects of fChick)
MM1_lmer <- lmer(Beak ~ Age + (1 | fNest), data = WS2)
summary(MM1_lmer)

MM1_lme <- lme(Beak ~ Age, random = ~ 1 | fNest, data = WS2)
summary(MM1_lme)

# Model 2 - Correct model: Two-way nested model with fNest and fChick
MM2_lmer <- lmer(Beak ~ Age + (1 | fNest / fChick), data = WS2)
summary(MM2_lmer)

WS2$fNest_fChick <- interaction(WS2$fNest, WS2$fChick, drop = TRUE)
MM2_lme <- lme(Beak ~ Age, random = list(fNest = ~1, fNest_fChick = ~1), data = WS2)
summary(MM2_lme)

## Compare models: lower AIC indicates better model fit
# AIC
AIC(MM1_lmer, MM2_lmer)
AIC(MM1_lme, MM2_lme)
```



LINEAR MIXED MODELS - EXAMPLE

- Simple LMM with one random factor fNest
- Two approaches using *lmer()* from *lme4* package and *lme()* from *nlme* package
- Results: Identical for both fixed effects (e.g., estimates) and random effects (standard deviation of intercepts and residuals)

```
> MM1_lmer <- lmer(Beak ~ Age + (1 | fNest), data = WS2)
> summary(MM1_lmer)
Linear mixed model fit by REML ['lmerMod']
Formula: Beak ~ Age + (1 | fNest)
Data: WS2
```

REML criterion at convergence: 10234.3

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-4.9103	-0.5406	-0.0259	0.5768	6.3796

Random effects:

Groups	Name	Variance	Std.Dev.
fNest	(Intercept)	57.23	7.565
	Residual	62.50	7.905

Number of obs: 1438, groups: fNest, 73

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	44.4417	0.9651	46.05
Age	2.9925	0.0169	177.08

Correlation of Fixed Effects:

	(Intr)
Age	-0.301

```
> MM1_lme <- lme(Beak ~ Age, random = ~ 1 | fNest, data = WS2)
> summary(MM1_lme)
Linear mixed-effects model fit by REML
Data: WS2
      AIC      BIC    logLik
10242.25 10263.33 -5117.127
```

Random effects:

Formula: ~1 | fNest
(Intercept) Residual
StdDev: 7.565109 7.905416

Fixed effects: Beak ~ Age

	Value	Std.Error	DF	t-value	p-value
(Intercept)	44.44171	0.9650576	1364	46.05084	0
Age	2.99246	0.0168990	1364	177.07929	0

Correlation:

(Intr)
Age -0.301

Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-4.91034447	-0.54059746	-0.02591983	0.57683245	6.37961831

Number of Observations: 1438

Number of Groups: 73

LINEAR MIXED MODELS - EXAMPLE

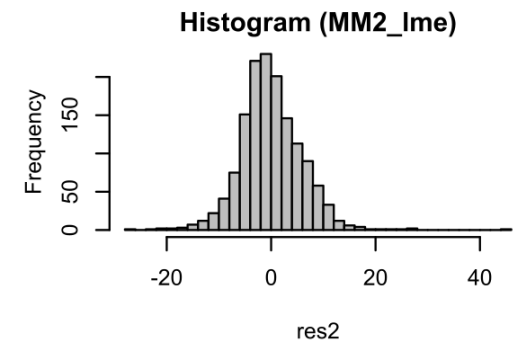
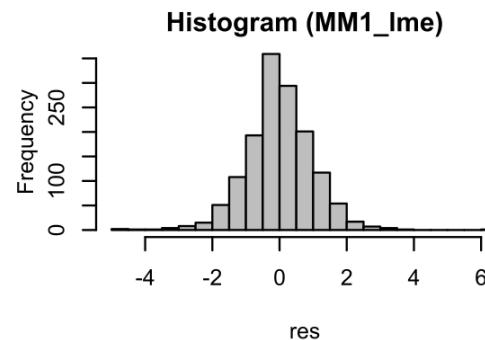
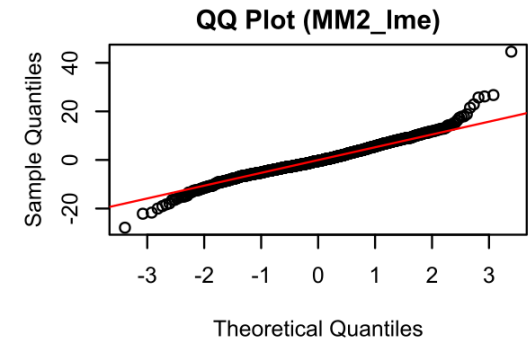
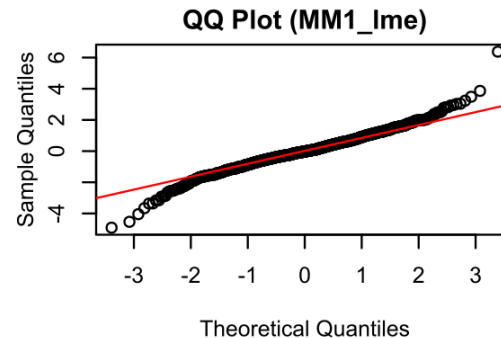
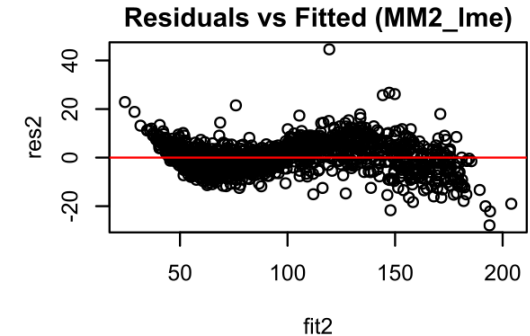
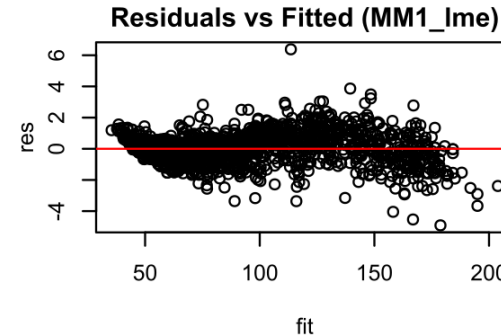
- LMM with random intercept only (1 | fNest) versus with nested random effect (1 | fNest / fChick)
- The more complex nested model provides a better solution: lower AIC, BIC, higher log-likelihood, and significant p-value

```
> AIC(MM1_lme, MM2_lme)
```

	df	AIC
MM1_lme	4	10242.254
MM2_lme	5	9942.592

```
> anova(MM1_lme, MM2_lme)
```

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
MM1_lme	1	4	10242.254	10263.33	-5117.127			
MM2_lme	2	5	9942.592	9968.94	-4966.296	1 vs 2	301.6619	<.0001



SUMMARY OF THE LECTURE

- Real-world data and experiments can be complex
 - Hierarchical/nested data structures
 - Different data distribution families and link
 - Fixed and random effects in models
- Decide sample size, replicates, and statistical approaches before conducting study
- Analyze data structure and assumptions to identify most suitable models to address research questions

FURTHER READINGS

- Hurlbert (1984). “Pseudoreplication and the design of ecological field experiments”
- Zuur et al. (2010) “A protocol for data exploration to avoid common statistical problems”. *Methods in Ecology and Evolution*, 1: 3-14. <https://doi.org/10.1111/j.2041-210X.2009.00001.x>
- Zuur et al. (2009) “Mixed Effects Models and Extensions in Ecology with R”
- Bolker (2015). “Linear and generalized linear mixed models”
- Brooks et al (2017). “glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling”



For further questions: Discussion forum on Moodle

